Anand Nagaraj

Digital Design Engineer · RTL · RISC-V · SoC Interconnects · Memory

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PROFILE

M.Sc. student in Design and Test of Integrated Circuits at TU Chemnitz with **3.5 years of industry experience** designing synthesizable RTL: RISC-V cores and custom instruction extensions, Network-on-Chip routers and a PCIe transaction-layer IP – from micro-architecture through verification to FPGA prototyping and open-source RTL-to-GDSII. Seeking a **working student position (up to 20 h/week) or internship** in digital, processor or memory design.

PROFESSIONAL EXPERIENCE

RTL Design & Verification Engineer (founding team) 11/2024 – 02/2026
Semiconductor IP start-up Bengaluru, India

- Designed synthesizable SystemVerilog RTL for **2D-mesh Network-on-Chip routers**: virtual-channel allocators, round-robin arbiters, multi-flit buffers, crossbar switches and routing logic.
- Extended the open-source **Ibex RISC-V core** with custom instructions and dedicated execution logic; verified against the Spike ISS and prototyped on FPGA.
- Added SVA on asynchronous-FIFO CDC boundaries, eliminating transaction deadlocks under heavy synthetic traffic.
- Built SystemVerilog / cocotb testbenches and a Python regression framework; mentored junior engineers and interns through the flow to GDSII.

RTL Design & Verification Engineer 09/2022 – 10/2024
Hardware engineering services company Kochi / Bengaluru, India

- Designed synthesizable Verilog RTL for a **PCIe Transaction Layer IP**: 32/64-bit request and completion TLP generation and parsing, packet framing, buffer management.
- Implemented credit-based flow control and end-to-end data integrity (32-bit ECRC generation/checking, poisoned-TLP detection).
- Synthesized blocks with Synopsys Design Compiler and closed setup/hold timing; brought up and debugged the IP on Xilinx FPGAs (Vivado ILA/VIO).
- Verified the IP with modular UVM testbenches and cocotb co-simulation (TLP ordering, protocol corner cases).

KEY PROJECTS

Moola-V (MR5) – 5-stage pipelined RV32IM RISC-V core
SystemVerilog · cocotb · Spike · RISCOF

- In-order 5-stage pipeline with full RAW forwarding (EX→EX, MEM→EX), cycle-accurate load-use stalls and branch/jump flush.
- Single-cycle multiplier and 32-cycle iterative radix-2 restoring divider in the datapath.
- **100 % compliance on all 49 RISCOF architectural test suites**; lock-step cocotb co-simulation against the Spike ISS. *github.com/AnandNag003/moola_riscv_core*

Post-quantum TLS retrofit on a RISC-V board – NextSEMI 2026 hackathon
C · OpenSSL 3.5 · ML-KEM · ML-DSA

- Retrofitted a TLS 1.3 client on an embedded RISC-V (RV64GC) Linux board with hybrid key exchange (X25519 + ML-KEM-768) and hybrid authentication (ECDSA P-256 + ML-DSA-44).
- Offset the post-quantum overhead with RISC-V bit-manipulation (Zba/Zbb) builds, multithreaded signature verification and CPU-frequency tuning: handshake **31 % faster than the legacy baseline** (14.5 vs 21.0 ms, 1,000 runs, own test server).

Dual-port synchronous SRAM macro – RTL to GDSII
Verilog · SkyWater 130 nm · OpenLane 2 · OpenROAD

- Parameterised 16 × 8 dual-port SRAM with byte write-masking and concurrent read/write collision arbitration (Tiny Tapeout).
- Full open-source flow RTL → GDSII: **DRC/LVS clean, STA sign-off**; 6T bit-cell sizing and SNM analysis in SPICE. *github.com/AnandNag003/tt_moola_dp_sram*

AMBA AHB-to-APB bridge
Verilog · SystemVerilog · UVM

- Cycle-accurate FSM translating pipelined AHB transfers into APB cycles; address decoding, HREADYOUT wait states, error mapping.
- UVM environment (AHB master / APB slave agents), constrained-random sequences and SVA – full coverage closure.

Mixed-signal frequency measurement system (TU Chemnitz)
VHDL · VHDL-AMS

- Dual-clock measurement unit with cooperating FSMs, double-flop synchronisers and request/acknowledge CDC handshakes.
- VHDL-AMS model of a MEMS sensor front-end (RC filter, VCO) with threshold logic and 115200-baud UART for end-to-end verification.

EDUCATION

M.Sc. Design and Test of Integrated Circuits 04/2026 – present
Technische Universität Chemnitz Chemnitz, Germany

- Modules: Verification of Digital Systems, Integrated Circuit Design (transistor level), Design of Heterogeneous Systems, Advanced IC Technology, Test and Design of Mixed-Signal Systems.
- **Deutschlandstipendium** 2026/27 – national merit scholarship (industry-sponsored).

B.E. Electronics & Telecommunication Engineering 08/2018 – 04/2022
Visvesvaraya Technological University (BMSIT) Bengaluru, India

- CGPA 8.51 / 10 (German grade ≈ 1.8). **Best Outgoing Student** of the department; state merit scholarship (100 % tuition waiver).
- Modules: CMOS VLSI Design, Computer Architecture, Digital System Design, Microprocessors.

Advanced VLSI Design & Verification – 6-month full-time programme 04/2022 – 09/2022
VLSI training institute Bengaluru, India

- SystemVerilog, UVM testbench architecture, SVA, constrained-random verification, coverage closure, AMBA AXI / AHB / APB, QuestaSim.

TECHNICAL SKILLS

Design	SystemVerilog, Verilog, VHDL, VHDL-AMS; FSMs, pipelining, arbitration, clock-domain crossing
Architectures	RISC-V (RV32IM, Ibex, custom extensions), Network-on-Chip, SRAM / memory subsystems, PCIe Transaction Layer, AMBA AXI4 / AHB / APB
Implementation	Synopsys Design Compiler, STA / timing closure, OpenLane 2, OpenROAD, SkyWater 130 nm, KLayout, Magic
FPGA	Xilinx Vivado, ILA / VIO on-chip debug, FPGA prototyping
Verification	UVM, SVA, cocotb, RISCOF, Spike ISS; QuestaSim, Xcelium, VCS, Verilator
Scripting	Python, Bash, C / C++, Linux, Git

PUBLICATION

A. Nagaraj et al., “Peripheral Shared Memory Over PCI Express Link,” in *Proc. 7th IEEE International Conference on Innovative Data Communication Technologies and Application (ICIDCA)*, 2025. – Shared-memory throughput, buffer sizing and latency over PCIe.

HONOURS & SELECTED ACTIVITIES

- **NextSEMI 2026 semiconductor hackathon** (SEMICON Europa × TUM Venture Labs, Munich, 11/2026) – one of 25 participants selected across Europe; post-quantum TLS project (see Key Projects).
- **Deutschlandstipendium** 2026/27, TU Chemnitz.
- **Best Outgoing Student** 2022 and state merit scholarship (100 % tuition waiver), VTU / BMSIT.

LANGUAGES

English – fluent (C1) | German – A2 (B1 course in progress) | Kannada – native | Hindi – fluent

FURTHER ACTIVITIES & INTERESTS

Volunteer, IEEE Student Branch TU Chemnitz (technical workshops, semiconductor seminars). Interests: distance running, cricket, hiking in Saxony.

Chemnitz, November 2026